

MSI5001– Introduction to AI: Concepts, Applications, and Evaluation

Semester I, 2025-26

Course Project Handout

- **Update:** Report page limit (29 September).

Project Deliverables & Key Dates

Each team is required to submit the project through the Coursemology submission folder (only one submission per group is allowed).

1.1 Deadlines

- Project proposal submission¹ – **1st October 2025, 10 pm**
- Final project submission – **4th November 2025, 10 pm**
- Project presentations (via Zoom) - **8th November 2025**

1.2 Submission Requirements

Your final submission must include the following:

- (a) **Code folder:** Provide your implementation in a Colab or Jupyter notebook. Ensure code is well-documented and clearly commented. This folder should contain the notebook and any additional codes used.
- (b) **Readme.pdf:** Include detailed instructions for running your project. The grader will follow these steps to execute your code successfully.
- (c) **Project Report (PDF):** A concise, (2+2)-page write-up describing your work. See Section 2.3 for detailed guidelines.
- (d) **Supporting materials:** Place all other relevant project documents here.

All items should be organized into a single folder, compressed into a .zip archive, and submitted on Coursemology. Use the following naming format: MSI5001_Team<number>_Project.zip. For example: MSI5001_Team5_Project.zip. Team numbers will be provided later.

¹<https://forms.cloud.microsoft/r/MXwpQRHjEC>

1.3 Presentation Slides and Zoom Link

Presentations will be held on Zoom. All team members must turn on their cameras. The Zoom link will be shared closer to the date. After presenting, one team member must upload the slides to Coursemology separately. Slides should be named in this format:

MSI5001_Team<number>_Presentation.pptx.

For example, MSI5001_Team5_Presentation.pptx.

Grading Criteria

The project is graded out of 100 marks, but only comments and final letter grades (derived from your score) will be released. The grading scheme, report format, and presentation requirements are outlined below:

2.1 Technical Implementation – 30%

This criteria evaluates the quality, correctness, and clarity of your AI model and data workflow.

1. Clean Data Pipeline: Demonstrates proper preprocessing, including handling of missing values, encoding of categorical variables, feature scaling, and effective feature selection or engineering.
2. Appropriate Model Choice: Selects AI models that are well-suited to the dataset and problem type.
3. Baseline Model Comparison: Includes a simple, meaningful baseline model (e.g., majority class predictor or mean regressor) for benchmarking performance.
4. Readability & Reproducibility: Code is well-organized, commented, and easy to follow. The notebook (e.g., Colab/Jupyter) should allow another person (including the grader) to run and understand the workflow without needing clarification.
5. Creativity / Customization: Shows thoughtful adaptation of models or pipeline design to improve performance, robustness, or domain relevance.

2.2 Evaluation & Insight – 20%

This criteria assesses how well the model's performance is measured, interpreted, and connected back to the problem context.

1. Use of Appropriate Metrics: Evaluation metrics are correctly chosen based on the machine learning task, and align with the project's goals.
2. Clarity of Evaluation Process: The evaluation strategy (e.g., train/test split, cross-validation) is clearly explained and correctly implemented.

3. **Baseline Comparison:** Results are compared against a meaningful baseline, showing how the chosen model performs relative to a simple benchmark.
4. **Interpretation of Results:** Results are clearly explained in relation to the original problem, highlighting key takeaways, surprising findings, or practical implications.
5. **Limitations & Error Analysis (Encouraged):** Discussion includes possible reasons for poor performance, sources of error, or constraints due to data quality, model assumptions, or overfitting.

2.3 Report & Presentation – 30%

This component evaluates how clearly and effectively your work is communicated, both in writing and during the final presentation. The report must adhere to the specified format provided here².

2.3.1 Report – 15%

- **Structure:** Report follows a clear and logical format — Introduction, Methodology, Results, Insights, Conclusion. The main report should be 2 pages, with additional 2 pages for appendix to add references, additional figures, tables, etc.
- **Language:** Writing is concise, coherent, and professional. Follows the prescribed formatting guidelines (e.g., font size, spacing, page limit).
- **Visuals:** Tables, charts, and diagrams are used effectively to present results and support key points.
- **References:** All external resources — including datasets, tools, academic works, or conceptual inspiration — are properly cited.

2.3.2 Presentation – 15%

- **Slide Design:** Slides are visually clean, with readable fonts, meaningful headings, and minimal clutter. Visuals aid understanding.
- **Flow:** Problem → Data → Model → Results → Insights. Your presentation should follow a clear and logical flow, starting from the problem definition, followed by an overview of the dataset, explanation of the model(s) used, presentation of the results, and concluding with key insights or takeaways.
- **Delivery:** All team members actively participate. Presenters speak clearly, stay within time limits, and demonstrate strong grasp of the content.

²<https://drive.google.com/drive/folders/1jq-wuryVbzNn5WcuXRFwxkJGWPgzEr91?usp=sharing>

- Q&A: The team responds questions with clarity, thoughtfulness, and evidence-based reasoning.

2.4 Team Contribution & Reflection – 20%

This criteria evaluates how effectively the team collaborated and how each member engaged with the project and learning process.

1. **Balanced Contribution:** Tasks are fairly and clearly distributed among team members (e.g., data preparation, model implementation, writing, presentation slides).
2. **Evidence of Teamwork:** Demonstrates consistent collaboration, communication, and mutual support throughout the project. This will be assessed based on the shared Google Doc link submitted in the project proposal form, which should document team discussions, task distribution, and project progress.
3. **Peer Assessment (to be submitted individually):** Each member may rate other members' contributions to ensure fairness. These ratings may be used to adjust individual scores if significant imbalance is reported.
4. **Individual Reflection:** Each member submits a short paragraph addressing:
 - What they personally contributed to the project
 - What they learned during the process
 - What challenges they faced and how they overcame them
5. Each member must submit their peer evaluation and individual reflection using this form³.

Additional Guidelines

- **Citing External Sources:** Acknowledge all external resources, including code, AI-generated content, or reference material, in the report, code, or slides. Failure to do so may incur penalties.
- **Tracking Contributions:** Use collaborative platforms (e.g., Google Docs, Slides) to document contributions; this may be used for dispute resolution (if any).
- **Planning Workload:** Agree early on workload distribution to ensure smooth collaboration.
- **Comprehension Over Execution:** Not everyone must code, but all members should understand the project and be able to explain it clearly.
- **Responsible AI Tool Use:** Tools like ChatGPT or GitHub Copilot may assist your work, but usage must be declared. You will be evaluated on understanding and application of concepts, not the tool's output.

³<https://forms.office.com/r/38bFxyNbAQ>

Datasets

Here, we provide some datasets that are curated by MSI5001 project coordinators. You will choose one of the dataset from the given list. We divide the datasets into several categories to make it easier for you to browse and select. Each dataset is accompanied by a README file that provides detailed information about the dataset. To help you select a dataset, we've provided an estimated difficulty level for each task, categorized as **Easy**, **Medium**, or **Challenging**:

- **Easy**: Datasets with well arranged tabular features. There is no (or small) additional learning required for you to work on this dataset – it's likely that you only need to apply some knowledge of what you've learned in this course.
- **Medium**: Text, image, or multi-table tabular dataset. To perform well, some preliminary analysis and a literature review may be required to understand potential issues and determine the best approach for utilizing the data.
- **Challenging**: Moderately complex image, text, or audio data. Additional learning is necessary to effectively use this dataset with machine learning models. In some cases, further analysis may also be required. To achieve optimal performance, reviewing a few research papers and applying recent machine learning techniques can be beneficial.

Naturally, the higher the difficulty, the more learning opportunities arise. This offers greater room for improvement and exploration, which creates additional value in your work.

3.1 Tabular Datasets

3.1.1 HDB Rent Price [**Easy**]

Rent prices in Singapore for the past 5 years

3.1.2 Bank Telemarketing Success [**Easy**]

Predictions of whether the client will subscribe to a term deposit (variable y) after advertisement through telemarketing.

3.1.3 Fraud in Electricity and Gas Consumption [**Medium**]

Fraudulent activities data from an electricity and gas company; containing information about client data and billing history from early 2000.

3.2 Textual Datasets

3.2.1 Book Genre Prediction [Medium]

Classifying each book into a genre based on its synopsis.

3.2.2 mRNA Classification [Medium]

A large RNA sequence dataset containing messenger RNA (mRNA) and other types of RNAs.

3.3 Clustering Datasets

3.3.1 Airline Sentiment [Challenging]

Clustering twitter posts about airline service based on their sentiment.

3.4 Time-series Datasets

3.4.1 Transformer Oil Temperature Prediction [Challenging]

Prediction of oil temperature of an electric transformer based on demand. Oil temperature serves as a good proxy to the electric transformer condition.

3.5 Image Datasets

3.5.1 Electrical Component Identification [Medium]

Identification of electrical components in RGB images.

3.5.2 Rice Leaf Disease [Easy]

Dataset containing RGB leaf images of healthy and diseased plants.

3.6 Audio Datasets

3.6.1 Animal Sound [Medium]

Animal dataset containing sounds of ten different animals.

3.6.2 Urban Sound [Medium]

An audio classification dataset consisting of several thousand samples of 10 types of urban sound.